

A Sharp Uniform-in-Time Wasserstein-1 Bound for Stochastic Runge–Kutta Integrators of Langevin Dynamics

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We consider two stochastic Runge–Kutta integrators proposed by Yang & Wang (2026) for overdamped Langevin dynamics whose potential is strongly convex outside a bounded region. In the companion paper (Ye, 2026) the law of the numerical solution was shown to approach the law of the exact solution at second order in Wasserstein-1 distance, uniformly in the number of steps, up to a logarithm of the step size. We remove the logarithm, and a Gaussian example shows that the second order is attained by the bias of the invariant law of the integrators, and that the squared bias of their time averages attains the order h^4 of the mean square error bound of the companion paper.

Keywords: overdamped Langevin dynamics; stochastic Runge–Kutta integrator; Wasserstein distance; Fisher information; uniform-in-time error bound; sharpness.

1. Introduction

The numerical integration of the Langevin dynamics plays a central role in sampling problems (Lelièvre & Stoltz, 2016) and in score-based diffusion models (Song et al., 2021). We consider the overdamped Langevin dynamics

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t, \quad X_0 = x \in \mathbb{R}^d, \quad (1.1)$$

where $(B_t)_{t \geq 0}$ is a standard Brownian motion in $\mathbb{R}^d$, and $U(x)$ is a given potential on $\mathbb{R}^d$. The Langevin dynamics (1.1) leaves the target distribution $\pi(x) \propto e^{-U(x)}$ invariant. A numerical solution to (1.1) is a discrete Markov chain $(Z_k)_{k \geq 0}$, where Z_k approximates X_{kh} at step size $h > 0$, and for a test function $f(x)$, writing $\pi(f) := \int_{\mathbb{R}^d} f d\pi$, the mean square error (MSE) of the time average over N steps is

$$\text{MSE}(N, h) := \mathbb{E} \left[\left| \frac{1}{N} \sum_{k=0}^{N-1} f(Z_k) - \pi(f) \right|^2 \right]. \quad (1.2)$$

Yang & Wang (2026) proposed, for (1.1), two stochastic Runge–Kutta (SRK) integrators that evaluate ∇U and no higher derivative. They established the strong order $\frac{3}{2}$ of both integrators. In the companion paper (Ye, 2026), the second author proved for these integrators, under a potential that is strongly convex outside a bounded region, the mean square error bound $\text{MSE}(N, h) = \mathcal{O}(\frac{1}{Nh} + h^4)$ for test functions with bounded derivatives up to third order, and the uniform-in-time Wasserstein-1 bound

$$\sup_{N \geq 1} \mathcal{W}_1(\text{Law}(Z_N), \text{Law}(X_{Nh})) \leq C(1 + |Z_0|)^6 h^2 \left(1 + \log \frac{1}{h}\right).$$

The additional logarithm comes from the proof *per se*. The local weak error of one step is read through the test function propagated by the semigroup of (1.1); the third derivative of the propagated

function is bounded by t^{-1} after time t when the test function is Lipschitz, and summing $h^3 t^{-1}$ over the steps with propagation time $t \leq 1$ produces $h^2 \log \frac{1}{h}$. For the random splitting Langevin Monte Carlo, under the same strong convexity outside a bounded region, Li et al. (2025) prove a Wasserstein-1 bound of order h^2 without a logarithm, by a local error in relative entropy that requires a regular initial density; Theorem 1 starts from a deterministic state and obtains the regularity it needs along the chain.

This paper removes the logarithm and shows that the order h^2 is sharp. Under the same conditions of Ye (2026), Theorem 1 proves the uniform-in-time Wasserstein-1 bound

$$\sup_{N \geq 1} \mathcal{W}_1(\text{Law}(Z_N), \text{Law}(X_{Nh})) \leq C(1 + |Z_0|)^6 h^2.$$

without the logarithm factor. Theorem 2 takes $U(x) = \frac{1}{2}x^2$ with $d = 1$ and verifies both SRK integrators reduce to the same Gaussian recursion. Moreover, the Wasserstein-1 distance between their invariant law and π is $\frac{1}{12}\sqrt{2/\pi}h^2$ to leading order, and the squared bias h^4 of the mean square error bound is attained by the test function $\cos x$.

The third derivative of the propagated test function enters only because the local weak error of one step is estimated pointwise in the state. Once the local weak error is averaged over the law μ_k of Z_k , one derivative can be moved from the test function onto μ_k by an integration by parts, at the price of the Fisher information $I(\mu_k)$ of μ_k . The integrators control that Fisher information themselves: one step of either integrator is a near-identity map of the state followed by an independent Gaussian of covariance $\frac{h}{2}I_d$, and the Gaussian part regularizes the law at the rate $I(\mu_k) \leq C(1 + (kh)^{-1})$ by Stam's inequality. The two singular factors that remain, $(kh)^{-1/2}$ from the Fisher information and $((N - k - 1)h)^{-1/2}$ from the second derivative of the propagated test function, sit at opposite ends of the chain, and their product is summable without a logarithm. The estimates of the exact dynamics and of one step of the integrators are those of Ye (2026).

Section 2 states the setting, the two SRK integrators and the main results. Section 3 recalls the results of the companion paper. Section 4 introduces the Fisher information technique and proves its two lemmas, Section 5 proves Theorem 1, and Section 6 proves Theorem 2.

2. Setting and main results

2.1. Notation and assumptions

We collect the notation and assumptions used in the paper. Fix a step size $h > 0$. The numerical solution $(Z_k)_{k \geq 0}$ of the overdamped Langevin dynamics (1.1) starts at $Z_0 = X_0$ and is driven by the same Brownian motion as the exact solution $(X_t)_{t \geq 0}$, so that Z_k approximates X_{kh} along a single trajectory. For each k , $(Z_k(t))_{t \geq 0}$ denotes the solution of (1.1) with $Z_k(0) = Z_k$, driven by the same Brownian motion as the step from Z_k to Z_{k+1} , so that $Z_k(h)$ is the exact step from Z_k , and $\mathbb{E}_k[\cdot] := \mathbb{E}[\cdot | Z_k]$ is the conditional expectation at step k . The law of Z_k is written μ_k and the law of X_{kh} is written ν_k ; the bounds below compare μ_N with ν_N as h decreases and with π as N grows. A measure and its density are written alike. The chain $(Z_k)_{k \geq 0}$ is Markov and the dynamics is time-homogeneous, so every statement about one step made at the step index 0 applies to the step from any Z_k after shifting the Brownian motion.

For probability measures μ, ν on $\mathbb{R}^d$, $\mathcal{W}_1$ denotes the Wasserstein-1 distance,

$$\mathcal{W}_1(\mu, \nu) = \sup_{f \in \text{Lip}_1(\mathbb{R}^d)} \left| \int_{\mathbb{R}^d} f(x) \mu(x) dx - \int_{\mathbb{R}^d} f(x) \nu(x) dx \right|, \quad (2.1)$$

where $\text{Lip}_1(\mathbb{R}^d)$ is the set of functions with Lipschitz constant at most 1.

Throughout, $\|\cdot\|$ denotes the operator norm of a tensor of any order greater than one,

$$\|A\| := \sup \{ |A[v_1, \dots, v_j]| : |v_1| = \dots = |v_j| = 1 \};$$

a matrix is a tensor of order two, and $A \succeq B$, or $B \preceq A$, means that $A - B$ is positive semidefinite.

For a vector v , $v^{\otimes j}$ is its j -fold tensor power, so $v^{\otimes 2} = vv^\top$. Derivatives of U contract with tensors: for a matrix Q and a vector v , $\nabla^3 U(x)[Q] \in \mathbb{R}^d$ and $\nabla^3 U(x)[v] \in \mathbb{R}^{d \times d}$ have entries $(\nabla^3 U[Q])_i = \sum_{j,k} \partial_i \partial_j \partial_k U Q_{jk}$ and $(\nabla^3 U[v])_{ij} = \sum_k \partial_i \partial_j \partial_k U v_k$, and $\nabla^4 U(x)[S] \in \mathbb{R}^d$ is defined likewise for a tensor S of order three. $\Delta U = \sum_i \partial_i^2 U$ is the Laplacian and $\nabla^2 \Delta U \in \mathbb{R}^{d \times d}$ its Hessian, $(\nabla^2 \Delta U)_{ij} = \sum_k \partial_i \partial_j \partial_k^2 U$. For matrices, $A : B := \text{tr}(A^\top B)$, so that $|A : B| \leq d \|A\| \|B\|$. For a function g , $\|\nabla g\|_\infty := \sup_x |\nabla g(x)|$, and likewise $\|\nabla^2 g\|_\infty$ and $\|\nabla^3 g\|_\infty$ with the operator norm.

Throughout, C denotes a positive constant depending only on the parameters m, M, R stated below and on the dimension d . It does not depend on the number of steps N or the step size h , and its value may change from one occurrence to the next. Likewise $h_0 > 0$ denotes a step size threshold depending only on m, M, R and d , whose value may change from one statement to the next. Derivatives of a function of the initial state x , such as the exact solution X_t or one step of the integrator, are written ∂_x , while ∇ is the gradient of a function on $\mathbb{R}^d$ such as U , a test function or a density. The error analysis runs under the following assumption.

Assumption 1 $U \in C^5(\mathbb{R}^d)$, and there is a constant $M > 0$ such that U satisfies

$$|\nabla U(0)| \leq M, \quad \max_{2 \leq i \leq 5} \|\nabla^i U(x)\| \leq M, \quad \forall x \in \mathbb{R}^d,$$

and there are constants $m > 0$ and $R \geq 0$ such that $\nabla^2 U(x) \succeq mI_d$ for every $|x| \geq R$.

Remark 1 (Convergence under slightly weaker assumptions) *The boundedness of $\nabla^i U$, $i = 3, 4, 5$ above, can be relaxed to the polynomial growth*

$$\|\nabla^i U(x)\| \leq A(1 + |x|)^r, \quad i = 3, 4, 5,$$

for some $A > 0$ and $r \geq 1$. The same result still holds after a few modifications to the proof. One key observation is that, the uniform-in-time moment bounds in Lemma 6 remain valid, mainly using the fact that the drift is still globally Lipschitz and far-field dissipative. Consequently, in the modified proof, the local-error and higher-variation estimates must instead carry polynomial weights, which are absorbed by higher moments in Lemma 8. The Hessian smoothing estimate in Lemma 3 can be recovered from the equation for $\nabla P_t g$ and a first-variation formula, using only the bounded Hessian. For the Fisher information estimate in Lemma 9, the uniform log-Jacobian estimate can be replaced by Gaussian integration by parts over a block of steps, conditional on the stage noises; the bound $\|\partial_x \mathcal{T}_{h,a} - I_d\|_\infty \leq Ch$ suffices to retain $I(\mu_k) \leq C(1 + (kh)^{-1})$. To simplify the presentation, we retain the stronger boundedness assumptions from Assumption 1 in the detailed proof below.

2.2. Stochastic Runge–Kutta integrators

We consider two SRK integrators proposed by [Yang & Wang \(2026\)](#). The first, which we name SRK-I, evaluates ∇U twice per step:

$$(I) : \quad \begin{cases} H_k = Z_k - \frac{3h}{4} \nabla U(Z_k) + \frac{3\sqrt{2}}{2h} \int_{kh}^{(k+1)h} (B_s - B_{kh}) ds, \\ Z_{k+1} = Z_k - \frac{h}{3} \nabla U(Z_k) - \frac{2h}{3} \nabla U(H_k) + \sqrt{2} (B_{(k+1)h} - B_{kh}). \end{cases} \quad (2.2)$$

The second, which we name SRK-II, evaluates ∇U three times per step. Both stages entering the update carry the weight $\frac{1}{2}$ and the noise coefficients $1 \pm \frac{1}{\sqrt{2}}$:

$$(II) : \quad \begin{cases} H_k^\pm = Z_k - \frac{h}{2} \nabla U(Z_k) + \frac{\sqrt{2}}{h} \left(1 \pm \frac{1}{\sqrt{2}}\right) \int_{kh}^{(k+1)h} (B_s - B_{kh}) ds, \\ Z_{k+1} = Z_k - \frac{h}{2} (\nabla U(H_k^+) + \nabla U(H_k^-)) + \sqrt{2} (B_{(k+1)h} - B_{kh}). \end{cases} \quad (2.3)$$

Both integrators are of strong order $\frac{3}{2}$ ([Yang & Wang, 2026](#)) and weak order 2 ([Ye, 2026](#)), the weak order measured against test functions with bounded derivatives up to third order; the Wasserstein-1 bound of Theorem 1 holds against Lipschitz test functions.

2.3. Main results

The proof of Theorem 1 sums the bounds of Section 4 along the chain, and the decay (3.5) of the exact dynamics extends the result from v_N to π .

Theorem 1 *Let Assumption 1 hold and let λ be the rate of Lemma 2. For every step size $h \in (0, h_0]$ and every $N \geq 1$,*

$$\mathcal{W}_1(\mu_N, v_N) \leq C(1 + |Z_0|)^6 h^2, \quad (2.4)$$

and

$$\mathcal{W}_1(\mu_N, \pi) \leq C(1 + |Z_0|) e^{-\lambda N h} + C(1 + |Z_0|)^6 h^2. \quad (2.5)$$

(Proof in Section 5.)

The order h^2 of Theorem 1 is tested on a Gaussian example.

Theorem 2 *Let $d = 1$ and $U(x) = \frac{1}{2}x^2$, which satisfies Assumption 1 with $m = M = 1$ and $R = 0$, so that $\pi = \mathcal{N}(0, 1)$. Then SRK-I and SRK-II coincide, and for every $h \in (0, 1]$ the chain $(Z_k)_{k \geq 0}$ has a*

Gaussian invariant law π_h , for which, as $h \rightarrow 0$,

$$\mathcal{W}_1(\pi_h, \pi) = \frac{1}{12} \sqrt{\frac{2}{\pi}} h^2 + \mathcal{O}(h^3). \quad (2.6)$$

For $f(x) = \cos x$ and every $Z_0 \in \mathbb{R}$, again as $h \rightarrow 0$,

$$\liminf_{N \rightarrow \infty} \text{MSE}(N, h) \geq \frac{e^{-1}}{144} h^4 + \mathcal{O}(h^5). \quad (2.7)$$

(Proof in Section 6.)

The pathwise coincidence of SRK-I and SRK-II is not particular to $U(x) = \frac{1}{2}x^2$: for every quadratic potential on $\mathbb{R}^d$, ∇U is affine, so each update depends on the stages only through ∇U at their weighted average, and that average is the same state $Z_k + D_{\Pi}$ for both integrators, with D_{Π} the common vector (3.14), the weighted average of the stage displacements.

Theorem 2 shows that the second order is optimal in Wasserstein-1 distance and in mean square error. In its setting the law of Z_N is Gaussian with mean and variance converging to those of π_h , so $\mu_N \rightarrow \pi_h$ in $\mathcal{W}_1$ as $N \rightarrow \infty$, by the coupling of two Gaussian laws through one standard Gaussian used in Section 6; letting $N \rightarrow \infty$ in (2.5) with $Z_0 = 0$ gives $\mathcal{W}_1(\pi_h, \pi) \leq Ch^2$ for $h \in (0, h_0]$, and (2.6) shows that this order is attained. Likewise the h^4 term of the mean square error bound $\mathcal{O}(\frac{1}{Nh} + h^4)$ of Ye (2026) is attained by (2.7).

3. Results from the companion paper

This section states the results of Ye (2026) that the proofs below use: the variation processes of the exact dynamics with their bounds, the contraction and smoothing properties of the dynamics, the notation of one step and the decomposition of the local error with its moment bounds, and, for the chain, its moment bound, the telescoping formula and the bound on the local weak error of one step. The statements are given in full, in the notation of Section 2, and their proofs are in Ye (2026). Every external result the paper uses is stated here, so the paper is self-contained except for the results imported from Ye (2026).

3.1. The exact dynamics

Write P_t for the semigroup of (1.1), $P_t g(x) := \mathbb{E}^x[g(X_t)]$, where $\mathbb{E}^x$ is the expectation for the solution started at $X_0 = x$. Let $J_t = \partial_x X_t \in \mathbb{R}^{d \times d}$ be the first variation process of the Langevin dynamics (1.1); it measures the sensitivity of $(X_t)_{t \geq 0}$ to its initial state $X_0 = x$. Differentiating (1.1) in x gives

$$\frac{dJ_t}{dt} = -\nabla^2 U(X_t) J_t, \quad J_0 = I_d. \quad (3.1)$$

Differentiating (3.1) in the initial direction $w \in \mathbb{R}^d$ defines the second variation $K_t[w] = \partial_w J_t \in \mathbb{R}^{d \times d}$, whose dynamics obeys the ordinary differential equation

$$\frac{dK_t[w]}{dt} = -\nabla^2 U(X_t) K_t[w] - (\nabla^3 U(X_t)[J_t w]) J_t, \quad K_0[w] = 0, \quad (3.2)$$

with the contraction $\nabla^3 U(x)[v] \in \mathbb{R}^{d \times d}$ of Section 2.1.

Lemma 1 Under Assumption 1, pathwise for all $t \geq 0$ and $w \in \mathbb{R}^d$,

$$\|J_t\| \leq e^{Mt}, \quad \|K_t[w]\| \leq Mt e^{2Mt} |w|.$$

Lemma 2 Under Assumption 1, there are constants $\lambda > 0$ and $C \geq 1$, depending only on m, M and R , such that for every $g \in C^1(\mathbb{R}^d)$ with $\|\nabla g\|_\infty < \infty$ and all $x, y \in \mathbb{R}^d$, $t \geq 0$,

$$|P_t g(x) - P_t g(y)| \leq C e^{-\lambda t} \|\nabla g\|_\infty |x - y|. \quad (3.3)$$

Lemma 2 is Corollary 2 of Eberle (2016), applied in Ye (2026) to the one-sided bound $\langle \nabla U(x) - \nabla U(y), x - y \rangle \geq -M|x - y|^2$, which Assumption 1 improves to $\frac{m}{2}|x - y|^2$ beyond a radius fixed by m, M and R . Taking $y \rightarrow x$ in (3.3) gives the gradient decay

$$\|\nabla P_t g\|_\infty \leq C e^{-\lambda t} \|\nabla g\|_\infty, \quad \forall t \geq 0. \quad (3.4)$$

Under Assumption 1 the first moment of π is finite, with a bound depending only on m, M, R and d , and $\pi(g) = \pi(P_t g)$ by invariance, so integrating (3.3) in y against π gives

$$|P_t g(x) - \pi(g)| \leq C e^{-\lambda t} \|\nabla g\|_\infty (1 + |x|), \quad \forall t \geq 0. \quad (3.5)$$

Bounded derivatives of $P_t g$ beyond the first come from the smoothing of the dynamics over a short time.

Lemma 3 Under Assumption 1, with λ the rate of Lemma 2 and C depending only on m, M, R and d , for every $g \in C^3(\mathbb{R}^d)$ with $\|\nabla^i g\|_\infty < \infty$ for $i = 1, 2, 3$ and every $t > 0$,

$$\|\nabla^2 P_t g\|_\infty \leq C \min(1, t)^{-1/2} e^{-\lambda t} \|\nabla g\|_\infty, \quad \|\nabla^3 P_t g\|_\infty \leq C \min(1, t)^{-1} e^{-\lambda t} \|\nabla g\|_\infty. \quad (3.6)$$

3.2. The local error of one step

Although the Wasserstein-1 error compares only the distribution laws, the proof needs the pathwise coupling: Lemma 10 differentiates the numerical and exact steps along a fixed Brownian path.

By coupling the same Brownian motion, the *local error* of one step is

$$\Delta Z_k := Z_{k+1} - Z_k(h), \quad (3.7)$$

the discrepancy between one numerical step and one exact step from a common state Z_k . The second quantity of the analysis is the product term

$$\Delta Q_k := \Delta Z_k (Z_{k+1} + Z_k(h) - 2Z_k)^\top. \quad (3.8)$$

Take the step index 0, with $(B_s)_{0 \leq s \leq h}$, $B_0 = 0$, the Brownian motion of the step; ΔZ_0 and ΔQ_0 are the local error (3.7) and the product (3.8) of that step, for whichever of the two integrators generates the step. This subsection splits ΔZ_0 into a *mean-zero* term $\mathcal{M}(\Delta Z_0)$, of zero conditional mean given Z_0 , and a *remainder* $\mathcal{R}(\Delta Z_0)$.

SRK-I, SRK-II and the exact step all carry the same noise increment $\sqrt{2}B_h$, so it cancels in the local error of both integrators:

$$\begin{aligned} \text{(I)} : \Delta Z_0 &= \int_0^h \nabla U(Z_0(s)) ds - \frac{h}{3} \nabla U(Z_0) - \frac{2h}{3} \nabla U(H_0), \\ \text{(II)} : \Delta Z_0 &= \int_0^h \nabla U(Z_0(s)) ds - \frac{h}{2} (\nabla U(H_0^+) + \nabla U(H_0^-)). \end{aligned} \quad (3.9)$$

The *stages* of an integrator are the states at which its update evaluates ∇U : for SRK-I, Z_0 with weight $\frac{1}{3}$ and H_0 of (2.2) with weight $\frac{2}{3}$; for SRK-II, H_0^+ and H_0^- of (2.3) with weight $\frac{1}{2}$ each. The step subtracts the *stage drift* Σ_0 , which is h times the weighted average of ∇U over the stages:

$$\begin{aligned} \text{(I)} : \Sigma_0 &= \frac{h}{3} \nabla U(Z_0) + \frac{2h}{3} \nabla U(H_0), \\ \text{(II)} : \Sigma_0 &= \frac{h}{2} (\nabla U(H_0^+) + \nabla U(H_0^-)), \end{aligned} \quad (3.10)$$

so that, by (3.9) and by the updates in (2.2) and (2.3),

$$\Delta Z_0 = \int_0^h \nabla U(Z_0(s)) ds - \Sigma_0, \quad Z_1 - Z_0 = \sqrt{2}B_h - \Sigma_0. \quad (3.11)$$

Define the Gaussian *stage noise* A_0 by

$$A_0 := \frac{\sqrt{2}}{h} \int_0^h B_s ds, \quad \text{Cov}(A_0) = \frac{2h}{3} I_d. \quad (3.12)$$

The displacements of the exact solution $Z_0(s)$ and of the stages $H_0, H_0^\pm$ from Z_0 are

$$\begin{aligned} W_s &= Z_0(s) - Z_0 = - \int_0^s \nabla U(Z_0(s')) ds' + \sqrt{2}B_s, \\ D_I &= H_0 - Z_0 = - \frac{3h}{4} \nabla U(Z_0) + \frac{3}{2} A_0, \\ D_{II}^\pm &= H_0^\pm - Z_0 = - \frac{h}{2} \nabla U(Z_0) + \left(1 \pm \frac{1}{\sqrt{2}}\right) A_0. \end{aligned} \quad (3.13)$$

Each *stage displacement* in (3.13) is a drift plus a multiple of A_0 . Averaged with the weights of (3.10), the stage displacements give the same vector for the two integrators, the stage Z_0 of SRK-I having displacement 0:

$$D_{II} := \frac{1}{2} (D_{II}^+ + D_{II}^-) = - \frac{h}{2} \nabla U(Z_0) + A_0 = \frac{2}{3} D_I. \quad (3.14)$$

This common vector is written D_{II} in the analysis of SRK-I as well. Across steps, let $\mathcal{F}_k := \sigma(Z_0, (B_s)_{s \leq kh})$ be the σ -algebra generated by the initial state and by the driving path up to time kh . Both Z_{k+1} and $Z_k(h)$ are functions of Z_k and of the increments $(B_s - B_{kh})_{kh \leq s \leq (k+1)h}$, so every one-step quantity is $\mathcal{F}_{k+1}$ -measurable. Those increments are independent of $\mathcal{F}_k$, so $\mathbb{E}[\cdot | \mathcal{F}_k] = \mathbb{E}_k[\cdot]$ for every one-step quantity.

From here on, $\|X\|_p := \mathbb{E}_0[|X|^p]^{1/p}$ for a random vector X and $p \geq 1$, with $|X|$ replaced by the operator norm $\|X\|$ when X is a random tensor of order two or more. Throughout this subsection, C depends only on M and d , as in Lemmas 4 and 5, and never on h or on Z_0 ; every exponent p occurring below is at most 4, so that the dependence of C on p is absorbed into its dependence on M and d . The following basic inequalities are used repeatedly:

$$\begin{aligned} \|B_s\|_p &\leq C\sqrt{s}, & \|\max_{s \leq h} B_s\|_p &\leq C\sqrt{h}, \\ \left\| \int_0^h B_s ds \right\|_p &\leq Ch^{3/2}, & |\nabla U(Z_0)| &\leq M(1 + |Z_0|). \end{aligned}$$

Every tensor contracted against $\nabla^3 U(Z_0)$ or $\nabla^4 U(Z_0)$ in the moment estimates below is a sum of tensor products of vectors, integrated over s or not, so the operator norm bound M of Assumption 1 bounds each summand by M times the product of the vector lengths. The matrix $\nabla^2 \Delta U(Z_0)$ of (3.21) is $\nabla^4 U(Z_0)$ contracted against I_d , so $\|\nabla^2 \Delta U\| \leq dM$.

For the displacement of the exact solution, the definition of W_s in (3.13) and $|\nabla U(Z_0(s'))| \leq |\nabla U(Z_0)| + M|W_{s'}|$ give, by Grönwall's inequality, the pathwise bound

$$|W_s| \leq e^{Mh} \left(s |\nabla U(Z_0)| + \sqrt{2} \max_{s' \leq h} |B_{s'}| \right), \quad 0 \leq s \leq h,$$

so for $h \leq 1$,

$$\|W_s\|_p \leq C\sqrt{h}(1 + |Z_0|), \quad \|W_s - \sqrt{2}B_s\|_p \leq Ch(1 + |Z_0|). \quad (3.15)$$

The same $\sqrt{h}$ bound holds for the stage displacements of (3.13), by the covariance of A_0 in (3.12), and for the step itself, by (3.11):

$$\begin{aligned} \|D\|_p &\leq C\sqrt{h}(1 + |Z_0|) \quad \text{for } D \in \{D_I, D_{II}^+, D_{II}^-\}, \\ \|Z_1 - Z_0\|_p &\leq C\sqrt{h}(1 + |Z_0|). \end{aligned} \quad (3.16)$$

Lemma 4 below writes $\mathcal{M}(\Delta Z_0)$ explicitly in integrals of the Brownian path.

Lemma 4 *Under Assumption 1, the local error (3.7) of SRK-I or SRK-II decomposes as*

$$\Delta Z_0 = \mathcal{M}(\Delta Z_0) + \mathcal{R}(\Delta Z_0), \quad (3.17)$$

where the mean-zero term is

$$\begin{aligned} \mathcal{M}(\Delta Z_0) &= -\sqrt{2}(\nabla^2 U(Z_0))^2 \int_0^h (h-s) B_s ds + \nabla^3 U(Z_0) \left[\int_0^h B_s^{\otimes 2} ds - \frac{3}{2h} \left(\int_0^h B_s ds \right)^{\otimes 2} \right] \\ &\quad - \sqrt{2} \nabla^3 U(Z_0) \left[\nabla U(Z_0) \otimes \int_0^h (s - c_1 h) B_s ds \right] + \frac{\sqrt{2}}{3} \nabla^4 U(Z_0) \left[\int_0^h B_s^{\otimes 3} ds - \frac{c_3}{h^2} \left(\int_0^h B_s ds \right)^{\otimes 3} \right], \end{aligned} \quad (3.18)$$

with $c_1 = \frac{3}{4}$, $c_3 = \frac{9}{4}$ for SRK-I and $c_1 = \frac{1}{2}$, $c_3 = \frac{5}{2}$ for SRK-II.

For $h \in (0, 1]$ there is a constant C , depending only on M and d , such that for all $Z_0 \in \mathbb{R}^d$,

$$\mathbb{E}_0[|\mathcal{M}(\Delta Z_0)|^4]^{1/4} \leq Ch^2(1 + |Z_0|), \quad \mathbb{E}_0[|\mathcal{R}(\Delta Z_0)|^4]^{1/4} \leq Ch^3(1 + |Z_0|)^4. \quad (3.19)$$

The product ΔQ_0 in (3.8) admits the same treatment: a mean-zero term $\mathcal{M}(\Delta Q_0)$ of order $h^{5/2}$ and a remainder $\mathcal{R}(\Delta Q_0)$ of order h^3 .

Lemma 5 *Under Assumption 1, the product (3.8) decomposes as*

$$\Delta Q_0 = \mathcal{M}(\Delta Q_0) + \mathcal{R}(\Delta Q_0), \quad (3.20)$$

where the mean-zero term is

$$\begin{aligned} \mathcal{M}(\Delta Q_0) = & 2\sqrt{2}\mathcal{M}(\Delta Z_0)B_h^\top + \frac{2h^3}{3}(\nabla^2 U(Z_0))^2 \\ & + 4h^3\left(\frac{1}{3} - \frac{c_1}{2}\right)\nabla^3 U(Z_0)[\nabla U(Z_0)] - \frac{4h^3}{3}\left(1 - \frac{c_3}{2}\right)\nabla^2 \Delta U(Z_0), \end{aligned} \quad (3.21)$$

with c_1, c_3 of Lemma 4.

For $h \in (0, 1]$, there is a constant C , depending only on M and d , such that for all $Z_0 \in \mathbb{R}^d$,

$$\mathbb{E}_0[\|\mathcal{M}(\Delta Q_0)\|^2]^{1/2} \leq Ch^{5/2}(1 + |Z_0|), \quad \mathbb{E}_0[\|\mathcal{R}(\Delta Q_0)\|^2]^{1/2} \leq Ch^3(1 + |Z_0|)^5. \quad (3.22)$$

By Minkowski's inequality, (3.19) bounds the local error itself: for $h \leq 1$,

$$\mathbb{E}_0[|\Delta Z_0|^4]^{1/4} \leq Ch^2(1 + |Z_0|)^4. \quad (3.23)$$

For a test function $G \in C^3(\mathbb{R}^d)$ with bounded derivatives up to third order, the weak error of one step is expanded about Z_k along the interpolated position

$$Y_\theta := \theta Z_{k+1} + (1 - \theta)Z_k(h), \quad \theta \in [0, 1], \quad (3.24)$$

as

$$G(Z_{k+1}) - G(Z_k(h)) = \langle \nabla G(Z_k), \Delta Z_k \rangle + \frac{1}{2} \nabla^2 G(Z_k) : \Delta Q_k + \rho_k^G, \quad (3.25)$$

with the remainder

$$\rho_k^G = \int_0^1 \int_0^1 (1-s) \left\langle \nabla^3 G(Z_k + s(Y_\theta - Z_k)) [(Y_\theta - Z_k)^{\otimes 2}], \Delta Z_k \right\rangle ds d\theta. \quad (3.26)$$

By taking the expectation of (3.25) conditioned on Z_k , the mean-zero terms of ΔZ_k and ΔQ_k drop out, and the remainders $\mathcal{R}(\Delta Z_k)$ and $\mathcal{R}(\Delta Q_k)$ of (3.17) and (3.20), together with ρ_k^G , are of order h^3 : this is the local weak error of one step, which Lemma 7 below bounds.

3.3. Stability and local weak error

Lemma 6 *Let Assumption 1 hold and let $q \geq 1$ be an integer. There are h_0 and C , depending on q as well, such that for every step size $h \in (0, h_0]$ and every $k \geq 0$,*

$$\begin{aligned} \mathbb{E}|Z_k|^{2q} & \leq C(1 + |Z_0|)^{2q}, \\ \max\{\mathbb{E}|Z_k(h) - Z_k|^{2q}, \mathbb{E}|Z_{k+1} - Z_k|^{2q}\} & \leq Ch^q(1 + |Z_0|)^{2q}. \end{aligned} \quad (3.27)$$

For $N \geq 1$ and $g \in C^1(\mathbb{R}^d)$ with a bounded gradient, the local weak errors of one step are summed along the chain by the telescoping formula

$$\mathbb{E}[g(Z_N)] - \mathbb{E}[g(X_{Nh})] = \sum_{k=0}^{N-1} \mathbb{E} \left[\mathbb{E}_k [G_k(Z_{k+1}) - G_k(Z_k(h)) \right], \quad (3.28)$$

where $G_k := P_{(N-k-1)h}g$ is the test function g propagated by the semigroup P_t of (1.1) over the time the dynamics still has to run after the step from Z_k , with $G_{N-1} = g$, so that the k -th summand is the local weak error of one step read through G_k . Each summand of (3.28) is bounded by the following lemma.

Lemma 7 *Let Assumption 1 hold, let $h \in (0, 1]$, and let $G \in C^3(\mathbb{R}^d)$ with $\|\nabla^i G\|_\infty < \infty$ for $i = 1, 2, 3$. Then, with C depending only on M and d , for every $k \geq 0$,*

$$\left| \mathbb{E}_k [G(Z_{k+1}) - G(Z_k(h))] \right| \leq Ch^3 (\|\nabla G\|_\infty + \|\nabla^2 G\|_\infty + \|\nabla^3 G\|_\infty) (1 + |Z_k|)^6. \quad (3.29)$$

4. Fisher information technique

This section replaces the pointwise estimate of Lemma 7, in which the third derivative of the propagated test function produces the logarithm in the bound of Ye (2026), by an estimate of the local weak error averaged over the law μ_k of Z_k .

Fix a smooth test function g with $\|\nabla g\|_\infty \leq 1$. The telescoping formula (3.28) writes the error between the laws of the numerical solution Z_N and the exact solution X_{Nh} as

$$\mathbb{E}[g(Z_N)] - \mathbb{E}[g(X_{Nh})] = \sum_{k=0}^{N-1} \mathbb{E} \left[\mathbb{E}_k [G_k(Z_{k+1}) - G_k(Z_k(h)) \right], \quad G_k = P_{(N-k-1)h}g,$$

and the summand at step k is an average over μ_k . Lemma 7 bounds it through the derivatives of G_k up to third order. Another control of that average, one that spares the third derivative, comes from the Fisher information

$$I(\mu) := \int_{\mathbb{R}^d} |\nabla \log \mu(x)|^2 \mu(x) dx, \quad (4.1)$$

taken at $\mu = \mu_k$.

Lemma 8 *Let Assumption 1 hold and let $h \in (0, h_0]$. Let Z_0 be independent of the Brownian motion driving the step, with law μ_0 positive and C^1 , $I(\mu_0)$ of (4.1) finite and $\mathbb{E}(1 + |Z_0|)^{12} < \infty$, and let $G \in C^3(\mathbb{R}^d)$ with $\|\nabla^i G\|_\infty < \infty$ for $i = 1, 2, 3$. Then*

$$\left| \mathbb{E}[G(Z_1) - G(Z_0(h))] \right| \leq Ch^3 (\|\nabla G\|_\infty + \|\nabla^2 G\|_\infty) \left(1 + \sqrt{I(\mu_0)}\right) (\mathbb{E}(1 + |Z_0|)^{12})^{1/2}. \quad (4.2)$$

(Proof in Section 4.2.)

Compared with Lemma 7, the bound (4.2) carries $\|\nabla G\|_\infty$ and $\|\nabla^2 G\|_\infty$ alone: the third derivative of G enters the hypotheses but not the bound, and the price is regularity of the law μ_0 , together with its twelfth moment, whereas averaging (3.29) would ask only for the sixth. To apply Lemma 8 at the steps $1 \leq k \leq N-2$, it remains to bound the Fisher information $I(\mu_k)$, and this bound the

integrators provide themselves. One step of SRK-I and SRK-II is a near-identity map of the state followed by an independent Gaussian of covariance $\frac{h}{2}I_d$, the structure (4.6). The map raises Fisher information to at most $(1 + Ch)I(\mu_k) + Ch$ by (4.42), and the Gaussian lowers it through Stam's inequality (4.44) (Blachman, 1965; Stam, 1959), which adds $h/(2d)$ to the reciprocal Fisher information of the transported law.

Lemma 9 *Under Assumption 1, for every step size $h \in (0, h_0]$, every initial law of Z_0 and every $k \geq 1$, the law μ_k of Z_k is positive and C^∞ , and*

$$I(\mu_k) \leq C \left(1 + \frac{1}{kh}\right). \quad (4.3)$$

(Proof in Section 4.3.)

Lemmas 8 and 9, with the moment bound (3.27) of the chain, bound the summand at step k , for $1 \leq k \leq N-2$, by

$$C(1 + |Z_0|)^6 h^3 (1 + (kh)^{-1/2}) (1 + (ih)^{-1/2}) e^{-\lambda ih}, \quad i := N - k - 1, \quad (4.4)$$

the factor $(1 + (kh)^{-1/2})$ from (4.3) and the factor $(1 + (ih)^{-1/2}) e^{-\lambda ih}$ from the bounds (3.4) and (3.6) on ∇G_k and $\nabla^2 G_k$. The two singular factors of (4.4) sit at opposite ends of the chain and $\int_0^1 (t(1-t))^{-1/2} dt$ is finite, so the sum over k is of order h^2 , uniformly in N .

Section 4.1 bounds the derivatives of one step, and Sections 4.2 and 4.3 use those bounds to prove Lemmas 8 and 9. In the rest of this section, the step runs from $Z_0 = x$: Z_1 is the numerical step, $Z_0(h)$ the exact step and ΔZ_0 the local error, each a function of x and of the path; C depends only on M and d , and the two integrators enter only through the stage drift Σ_0 of (3.10).

4.1. The step as a map of the state

The stage noise A_0 and the increment $\sqrt{2}B_h$ are jointly Gaussian, with $\text{Cov}(\sqrt{2}B_h, A_0) = hI_d$ and the covariance of A_0 recorded in (3.12). Regressing $\sqrt{2}B_h$ on A_0 therefore gives

$$\sqrt{2}B_h = \frac{3}{2}A_0 + \sqrt{\frac{h}{2}}\eta_0, \quad \eta_0 \sim \mathcal{N}(0, I_d) \text{ independent of } A_0. \quad (4.5)$$

The randomness of B_h and $\int_0^h B_s ds$ is thus carried by the two independent Gaussian vectors A_0 and η_0 . By (3.13), the stage drift (3.10) is a function $\Sigma_0(x, a)$ of the state $Z_0 = x$ and the stage noise $A_0 = a$ alone, so (3.11) and (4.5) write one step as a map of the state followed by an independent Gaussian:

$$Z_1 = \mathcal{T}_{h, A_0}(Z_0) + \sqrt{\frac{h}{2}}\eta_0, \quad \mathcal{T}_{h, a}(x) := x - \Sigma_0(x, a) + \frac{3}{2}a, \quad (4.6)$$

where the stage drift function $\Sigma_0(x, a)$ is, for SRK-I and SRK-II,

$$\begin{aligned} \text{(I)} : \Sigma_0(x, a) &= \frac{h}{3} \nabla U(x) + \frac{2h}{3} \nabla U \left(\underbrace{x - \frac{3h}{4} \nabla U(x) + \frac{3}{2}a}_{D_I} \right), \\ \text{(II)} : \Sigma_0(x, a) &= \frac{h}{2} \nabla U \left(\underbrace{x - \frac{h}{2} \nabla U(x) + \left(1 + \frac{1}{\sqrt{2}}\right)a}_{D_{II}^+} \right) + \frac{h}{2} \nabla U \left(\underbrace{x - \frac{h}{2} \nabla U(x) + \left(1 - \frac{1}{\sqrt{2}}\right)a}_{D_{II}^-} \right). \end{aligned} \quad (4.7)$$

The map $\mathcal{T}_{h,a}$ is the *step map* of the integrator. The displacements D_1 and $D_{\Pi}^{\pm}$ also depend on x explicitly, through the gradient $\nabla U(x)$, and we drop this dependence for notational convenience.

Lemma 10 below establishes the regularity of the step map $\mathcal{T}_{h,a}$; its proof is tedious but standard in stochastic analysis. In particular, the constant C in Lemma 10 does not depend on h or a .

Lemma 10 *Under Assumption 1, with C depending only on M and d , the following hold for every $h \in (0, h_0]$. For every $a \in \mathbb{R}^d$, the map $\mathcal{T}_{h,a}$ is a C^2 diffeomorphism of $\mathbb{R}^d$ with*

$$\|\partial_x \mathcal{T}_{h,a} - I_d\|_{\infty} + \|\partial_x^2 \mathcal{T}_{h,a}\|_{\infty} + \|\nabla \log \det \partial_x \mathcal{T}_{h,a}\|_{\infty} \leq Ch. \quad (4.8)$$

Along every Brownian path, the maps $x \mapsto Z_1$ and $x \mapsto Z_0(h)$ are C^2 , with

$$\|\partial_x Z_1 - I_d\|_{\infty} + \|\partial_x Z_0(h) - I_d\|_{\infty} + \|\partial_x^2 Z_1\|_{\infty} + \|\partial_x^2 Z_0(h)\|_{\infty} \leq Ch. \quad (4.9)$$

The derivative of the local error satisfies, for every $x \in \mathbb{R}^d$,

$$\|\partial_x \Delta Z_0\|_2 \leq Ch^2(1 + |x|)^2. \quad (4.10)$$

Proof of Lemma 10 The proof is written for SRK-I; the changes for SRK-II are recorded at the end. Since $\partial_x \mathcal{T}_{h,a} = I_d - \partial_x \Sigma_0$ by (4.6), differentiating the form (I) of (4.7) in x , where the stage argument is $x + D_1$ with $\partial_x D_1 = -\frac{3h}{4} \nabla^2 U(x)$, gives the Jacobian difference

$$\partial_x \mathcal{T}_{h,a}(x) - I_d = -\frac{h}{3} \nabla^2 U(x) - \frac{2h}{3} \nabla^2 U(x + D_1) \left(I_d - \frac{3h}{4} \nabla^2 U(x) \right), \quad (4.11)$$

and one more derivative in the direction v gives

$$\begin{aligned} \partial_x^2 \mathcal{T}_{h,a}(x)[v] &= -\frac{h}{3} \nabla^3 U(x)[v] + \frac{h^2}{2} \nabla^2 U(x + D_1) \nabla^3 U(x)[v] \\ &\quad - \frac{2h}{3} \nabla^3 U(x + D_1) \left[\left(I_d - \frac{3h}{4} \nabla^2 U(x) \right) v \right] \left(I_d - \frac{3h}{4} \nabla^2 U(x) \right). \end{aligned} \quad (4.12)$$

Since $\|\nabla^2 U\|, \|\nabla^3 U\| \leq M$ and $h \leq 1$, (4.11) and (4.12) give

$$\|\partial_x \mathcal{T}_{h,a} - I_d\|_{\infty} \leq Mh \left(1 + \frac{Mh}{2} \right), \quad \|\partial_x^2 \mathcal{T}_{h,a}\|_{\infty} \leq Mh \left(1 + \frac{3Mh}{4} \right)^2 + \frac{M^2 h^2}{2},$$

both at most Ch . Let $h_0 \leq 1$ be such that the right sides of (4.8) and (4.9) are at most $\frac{1}{2}$ for $h \leq h_0$; h_0 depends only on M and d . Then $\mathcal{T}_{h,a} - \text{id}$ is Lipschitz with constant $\frac{1}{2}$, so $\mathcal{T}_{h,a}$ is injective, and for every y the map $x \mapsto y - (\mathcal{T}_{h,a} - \text{id})(x)$ is a contraction of constant $\frac{1}{2}$, whose fixed point solves $\mathcal{T}_{h,a}(x) = y$; hence $\mathcal{T}_{h,a}$ is a bijection. Its Jacobian is invertible at every point, so by the inverse function theorem the

inverse map is C^2 as well, and $\mathcal{T}_{h,a}$ is a C^2 diffeomorphism. Its Jacobian satisfies

$$\|(\partial_x \mathcal{T}_{h,a})^{-1}\| \leq (1 - \|\partial_x \mathcal{T}_{h,a} - I_d\|)^{-1} \leq 2,$$

and $\det \partial_x \mathcal{T}_{h,a}$, equal to 1 at $h = 0$ and nowhere zero, is positive. Finally, for every $v \in \mathbb{R}^d$,

$$|\nabla \log \det \partial_x \mathcal{T}_{h,a}(x) \cdot v| = |\text{tr}((\partial_x \mathcal{T}_{h,a})^{-1} \partial_x^2 \mathcal{T}_{h,a}[v])| \leq 2d \|\partial_x^2 \mathcal{T}_{h,a}\| |v| \leq Ch |v|.$$

By (4.6), Z_1 is the step map $\mathcal{T}_{h,A_0}$ evaluated at $Z_0 = x$ plus a noise term independent of x , so

$$\partial_x Z_1 = \partial_x \mathcal{T}_{h,A_0}, \quad \partial_x^2 Z_1 = \partial_x^2 \mathcal{T}_{h,A_0},$$

and the bounds on Z_1 in (4.9) are contained in (4.8). For the exact step, the noise of (1.1) is additive, so $Z_0(t) - \sqrt{2}B_t$ solves an ordinary differential equation with a C^4 vector field along every path, and $x \mapsto Z_0(h)$ is C^2 with $\partial_x Z_0(h) = J_h$ and $\partial_x^2 Z_0(h)[v] = K_h[v]$, the variations (3.1) and (3.2) along the path. Lemma 1 gives $\|K_h[v]\| \leq Mhe^{2Mh}|v|$, and integrating (3.1) with $\|J_s\| \leq e^{Ms}$ gives

$$\|J_h - I_d\| = \left\| \int_0^h \nabla^2 U(Z_0(s)) J_s ds \right\| \leq M \int_0^h e^{Ms} ds \leq Mhe^{Mh},$$

which proves the bounds on $Z_0(h)$ in (4.9).

For (4.10), differentiate the local error in (3.11):

$$\partial_x \Delta Z_0 = \int_0^h \nabla^2 U(Z_0(s)) J_s ds - \partial_x \Sigma_0. \quad (4.13)$$

The two terms on the right of (4.13), the integral from the exact step and $\partial_x \Sigma_0$ from the stages, are expanded about x to second order; they are called the exact side and the stage side below. Write

$$\mathcal{E}_1(v) := \nabla^2 U(x+v) - \nabla^2 U(x) - \nabla^3 U(x)[v] \quad (4.14)$$

for the Taylor remainder of $\nabla^2 U$ at a displacement v , which satisfies

$$\|\mathcal{E}_1(v)\| \leq \frac{M}{2} |v|^2. \quad (4.15)$$

On the exact side, with $Z_0(s) = x + W_s$ by (3.13), Taylor's formula gives

$$\nabla^2 U(Z_0(s)) = \nabla^2 U(x) + \nabla^3 U(x)[W_s] + \mathcal{E}_1(W_s). \quad (4.16)$$

The remainder $\mathcal{E}_2(s)$ is defined by the non-Gaussian part of the first variation process $J_t = \partial_x X_t$ of (3.1),

$$J_s = I_d - s \nabla^2 U(x) + \mathcal{E}_2(s). \quad (4.17)$$

Integrating (3.1), $\mathcal{E}_2(s)$ is expressed as

$$\mathcal{E}_2(s) = - \int_0^s \left((\nabla^2 U(Z_0(r)) - \nabla^2 U(x)) J_r + \nabla^2 U(x) (J_r - I_d) \right) dr,$$

and $\|\nabla^2 U(Z_0(r)) - \nabla^2 U(x)\| \leq M|W_r|, \|J_r\| \leq e^{Mr}, \|J_r - I_d\| \leq Mre^{Mr}$ with (3.15) give

$$\|\mathcal{E}_2(s)\|_4 \leq \int_0^s (Me^{Mr} \|W_r\|_4 + M^2 re^{Mr}) dr \leq Cs\sqrt{h}(1+|x|). \quad (4.18)$$

On the stage side, $\partial_x \Sigma_0 = I_d - \partial_x \mathcal{T}_{h,a}$ by (4.6), and (4.11) with (4.14) at $v = D_I$ gives

$$\partial_x \Sigma_0 = \frac{h}{3} \nabla^2 U(x) + \frac{2h}{3} \left(\nabla^2 U(x) + \nabla^3 U(x)[D_I] + \mathcal{E}_1(D_I) \right) \left(I_d - \frac{3h}{4} \nabla^2 U(x) \right). \quad (4.19)$$

When (4.16), (4.17) and (4.19) are multiplied out, the terms free of W_s , D_I , $\mathcal{E}_1$ and $\mathcal{E}_2$ agree on the two sides and cancel in the difference:

$$\begin{aligned} \int_0^h \nabla^2 U(x) (I_d - s \nabla^2 U(x)) ds &= \frac{h}{3} \nabla^2 U(x) + \frac{2h}{3} \nabla^2 U(x) \left(I_d - \frac{3h}{4} \nabla^2 U(x) \right) \\ &= h \nabla^2 U(x) - \frac{h^2}{2} (\nabla^2 U(x))^2. \end{aligned} \quad (4.20)$$

The remaining leading terms are $\nabla^3 U(x) [\int_0^h W_s ds]$ on the exact side and $\frac{2h}{3} \nabla^3 U(x)[D_I]$ on the stage side. The stage-side term equals $h \nabla^3 U(x)[D_{II}]$ by (3.14). The expression of D_{II} there and the definition (3.12) of A_0 write hD_{II} as

$$hD_{II} = \sqrt{2} \int_0^h B_s ds - \frac{h^2}{2} \nabla U(x). \quad (4.21)$$

Subtracting the stage side (4.19) from the exact side (4.16) and (4.17) in (4.13), with the cancellation (4.20) and the identity (4.21),

$$\partial_x \Delta Z_0 = \nabla^3 U(x) \left[\int_0^h (W_s - \sqrt{2} B_s) ds \right] + \frac{h^2}{2} \nabla^3 U(x) [\nabla U(x)] + \mathcal{E}, \quad (4.22)$$

where $\mathcal{E}$ collects the products of two small factors,

$$\begin{aligned} \mathcal{E} &:= \int_0^h \left(\mathcal{E}_1(W_s) J_s + (\nabla^2 U(x) + \nabla^3 U(x)[W_s]) \mathcal{E}_2(s) - s \nabla^3 U(x)[W_s] \nabla^2 U(x) \right) ds \\ &\quad - \frac{2h}{3} \mathcal{E}_1(D_I) \left(I_d - \frac{3h}{4} \nabla^2 U(x) \right) + \frac{h^2}{2} \nabla^3 U(x)[D_I] \nabla^2 U(x). \end{aligned} \quad (4.23)$$

The five terms of (4.23) are bounded in $\|\cdot\|_2$ by Hölder's inequality, the bound (4.15) on $\mathcal{E}_1$, the bound (4.18) on $\mathcal{E}_2(s)$, $\|J_s\| \leq e^{Ms}$, and (3.15) and (3.16) for W_s and D_I :

$$\begin{aligned} \|\mathcal{E}\|_2 &\leq \int_0^h \left(\frac{M}{2} e^{Ms} \|W_s\|_4^2 + M(1 + \|W_s\|_4) \|\mathcal{E}_2(s)\|_4 + sM^2 \|W_s\|_2 \right) ds \\ &\quad + \frac{2h}{3} \left(1 + \frac{3Mh}{4} \right) \frac{M}{2} \|D_I\|_4^2 + \frac{h^2}{2} M^2 \|D_I\|_2 \leq Ch^2(1+|x|)^2. \end{aligned} \quad (4.24)$$

The integral in (4.22) is bounded by (3.15) for $\|W_s - \sqrt{2} B_s\|_2$,

$$\left\| \int_0^h (W_s - \sqrt{2} B_s) ds \right\|_2 \leq \int_0^h \|W_s - \sqrt{2} B_s\|_2 ds \leq Ch^2(1+|x|). \quad (4.25)$$

With $\|\nabla^3 U\| \leq M$ and $|\nabla U(x)| \leq M(1 + |x|)$, the three terms of (4.22) are bounded, in this order, by (4.25), by these two bounds, and by (4.24):

$$\|\partial_x \Delta Z_0\|_2 \leq CMh^2(1 + |x|) + \frac{h^2}{2} M^2(1 + |x|) + Ch^2(1 + |x|)^2 \leq Ch^2(1 + |x|)^2,$$

which is (4.10). For SRK-II, differentiating the form (II) of (4.7), where the stage arguments are $x + D_{\Pi}^{\pm}$ with $\partial_x D_{\Pi}^{\pm} = -\frac{h}{2} \nabla^2 U(x)$, gives

$$\partial_x \mathcal{T}_{h,a}(x) - I_d = -\frac{h}{2} \sum_{\pm} \nabla^2 U(x + D_{\Pi}^{\pm}) \left(I_d - \frac{h}{2} \nabla^2 U(x) \right)$$

in place of (4.11). One more derivative gives the analogue of (4.12) with the two stages at weight $\frac{h}{2}$ each and $\frac{3h}{4}$ replaced by $\frac{h}{2}$, so that $\|\partial_x \mathcal{T}_{h,a} - I_d\|_{\infty} \leq Mh(1 + \frac{Mh}{2})$ and $\|\partial_x^2 \mathcal{T}_{h,a}\|_{\infty} \leq Mh(1 + \frac{Mh}{2})^2 + \frac{M^2 h^2}{2}$, both at most Ch ; hence (4.8) and (4.9) follow as for SRK-I. In place of (4.19), the stage side is

$$\partial_x \Sigma_0 = \frac{h}{2} \sum_{\pm} \left(\nabla^2 U(x) + \nabla^3 U(x) [D_{\Pi}^{\pm}] + \mathcal{E}_1(D_{\Pi}^{\pm}) \right) \left(I_d - \frac{h}{2} \nabla^2 U(x) \right). \quad (4.26)$$

The terms of (4.26) free of $D_{\Pi}^{\pm}$ and $\mathcal{E}_1$ sum to $h \nabla^2 U(x) - \frac{h^2}{2} (\nabla^2 U(x))^2$, the right side of (4.20). The leading term of the stage side is $\frac{h}{2} \nabla^3 U(x) [D_{\Pi}^+ + D_{\Pi}^-] = h \nabla^3 U(x) [D_{\Pi}]$ by (3.14), so (4.22) holds with the same leading terms. In (4.23), the term in $\mathcal{E}_1(D_1)$ is replaced by $-\frac{h}{2} \sum_{\pm} \mathcal{E}_1(D_{\Pi}^{\pm}) (I_d - \frac{h}{2} \nabla^2 U(x))$ and the term in $\nabla^3 U(x) [D_1]$ by $\frac{h^2}{4} \nabla^3 U(x) [D_{\Pi}^+ + D_{\Pi}^-] \nabla^2 U(x)$, which (3.16) bounds as in (4.24). $\square$

4.2. Proof of Lemma 8

The proof rests on an integration by parts against the initial law, which trades the third derivative $\nabla^3 G$ in the remainder ρ_0^G of (4.30) for the second derivative $\nabla^2 G$.

Proof of Lemma 8 Let h_0 be the threshold of Lemma 10. Conditioning on $Z_0 = x$ writes the local weak error in the left side of (4.2) as

$$\left| \int_{\mathbb{R}^d} \mathbb{E}_0 [G(Z_1) - G(Z_0(h))] \mu_0(x) dx \right|, \quad (4.27)$$

so the step from x is bounded first. Take $\mathbb{E}_0$ in (3.25) at $k = 0$ with $Z_0 = x$. By the decompositions (3.17) and (3.20), the mean-zero terms of ΔZ_0 and ΔQ_0 drop out under $\mathbb{E}_0$, and the bounds (3.19) and (3.22) on $\mathcal{R}(\Delta Z_0)$ and $\mathcal{R}(\Delta Q_0)$ give

$$\left| \mathbb{E}_0 [\langle \nabla G(x), \Delta Z_0 \rangle] \right| + \left| \mathbb{E}_0 \left[\frac{1}{2} \nabla^2 G(x) : \Delta Q_0 \right] \right| \leq Ch^3 (\|\nabla G\|_{\infty} + \|\nabla^2 G\|_{\infty}) (1 + |x|)^5. \quad (4.28)$$

Integrating (4.28) against μ_0 , with the Cauchy–Schwarz inequality applied to the moment, bounds the integral of its left side by

$$Ch^3 (\|\nabla G\|_{\infty} + \|\nabla^2 G\|_{\infty}) \int_{\mathbb{R}^d} (1 + |x|)^5 \mu_0(x) dx \leq Ch^3 (\|\nabla G\|_{\infty} + \|\nabla^2 G\|_{\infty}) (\mathbb{E}(1 + |Z_0|)^{12})^{1/2}. \quad (4.29)$$

It remains to bound the remainder ρ_0^G of (3.25), which is (3.26) at $k = 0$:

$$\rho_0^G = \int_0^1 \int_0^1 (1-s) \left\langle \nabla^3 G(x + s v_\theta) [v_\theta^{\otimes 2}], \Delta Z_0 \right\rangle ds d\theta, \quad (4.30)$$

where v_θ is the interpolated displacement from x , with the interpolated position Y_θ of (3.24),

$$v_\theta(x) := Y_\theta - x = \theta(Z_1 - x) + (1 - \theta)(Z_0(h) - x), \quad \theta \in [0, 1]. \quad (4.31)$$

Fix $s, \theta \in [0, 1]$ and the Brownian path, and read v_θ of (4.31) and ΔZ_0 as functions of x . By (4.9) and the choice of h_0 , v_θ and the Jacobian $I_d + s \partial_x v_\theta$ of $x \mapsto x + s v_\theta$ satisfy

$$\|\partial_x v_\theta\|_\infty + \|\partial_x^2 v_\theta\|_\infty \leq Ch \leq \frac{1}{2}, \quad \|(I_d + s \partial_x v_\theta)^{-1}\|_\infty \leq 2. \quad (4.32)$$

The third derivative of G in (4.30) is moved onto μ_0 by an integration by parts in x ; to that end set

$$w_{s,\theta}(x) := (I_d + s \partial_x v_\theta)^{-1} \Delta Z_0, \quad \Lambda_{s,\theta}(x) := \nabla^2 G(x + s v_\theta) [v_\theta, v_\theta]. \quad (4.33)$$

By (4.32), by the operator norm of $\nabla^2 G$, and by differentiating $w_{s,\theta}$ of (4.33), in this order,

$$\begin{aligned} |w_{s,\theta}| &\leq 2 |\Delta Z_0|, \quad |\Lambda_{s,\theta}| \leq \|\nabla^2 G\|_\infty |v_\theta|^2, \\ \|\partial_x w_{s,\theta}\| &\leq \|(I_d + s \partial_x v_\theta)^{-1}\| (\|\partial_x \Delta Z_0\| + s \|\partial_x^2 v_\theta\| |w_{s,\theta}|) \leq C (\|\partial_x \Delta Z_0\| + h |\Delta Z_0|). \end{aligned} \quad (4.34)$$

Differentiating $\Lambda_{s,\theta}$ of (4.33) gives

$$\nabla \Lambda_{s,\theta} = (I_d + s \partial_x v_\theta)^\top \nabla^3 G(x + s v_\theta) [v_\theta^{\otimes 2}] + 2(\partial_x v_\theta)^\top \nabla^2 G(x + s v_\theta) v_\theta, \quad (4.35)$$

and pairing (4.35) with $w_{s,\theta}$ turns the contraction in (4.30) into a derivative,

$$\left\langle \nabla^3 G(x + s v_\theta) [v_\theta^{\otimes 2}], \Delta Z_0 \right\rangle = w_{s,\theta} \cdot \nabla \Lambda_{s,\theta} - 2 \nabla^2 G(x + s v_\theta) [(\partial_x v_\theta) w_{s,\theta}, v_\theta], \quad (4.36)$$

in which $w_{s,\theta} \cdot \nabla \Lambda_{s,\theta}$ is the term that integration by parts against μ_0 turns into the two terms of (4.40). The moments of the objects involved are

$$\|v_\theta\|_4 \leq C\sqrt{h}(1 + |x|), \quad \|\Delta Z_0\|_4 \leq Ch^2(1 + |x|)^4, \quad \|\partial_x \Delta Z_0\|_2 \leq Ch^2(1 + |x|)^2, \quad (4.37)$$

the bound on v_θ by (3.15) and (3.16), those on ΔZ_0 and $\partial_x \Delta Z_0$ by (3.23) and (4.10). By Hölder's inequality with (4.37) and the bound on $\partial_x v_\theta$ in (4.32), uniformly in s and θ ,

$$\mathbb{E}_0[|v_\theta|^2 (\|\partial_x \Delta Z_0\| + h |\Delta Z_0|)] + \mathbb{E}_0[|v_\theta|^2 |\Delta Z_0|] + \mathbb{E}_0[\|\partial_x v_\theta\| |\Delta Z_0| |v_\theta|] \leq Ch^3(1 + |x|)^6. \quad (4.38)$$

The three moments of (4.38) control, in this order, the terms $\Lambda_{s,\theta} \mu_0 \operatorname{div} w_{s,\theta}$ and $\Lambda_{s,\theta} \nabla \mu_0 \cdot w_{s,\theta}$ of (4.40), and the term in $\nabla^2 G$ of (4.36).

The integration by parts is carried out with a cutoff $\chi_r \in C^1(\mathbb{R}^d)$, equal to 1 on $|x| \leq r$ and to 0 on $|x| \geq 2r$, with $|\nabla \chi_r| \leq C/r$; it is performed pathwise in (4.39), and (4.40) is its form after the expectation

over the path. The cutoff is needed because the integral is over all of $\mathbb{R}^d$ and $\mu_0 w_{s,\theta}$ is not assumed to decay at infinity. Since $\chi_r \mu_0 w_{s,\theta}$ is C^1 with compact support,

$$\begin{aligned} & \int_{\mathbb{R}^d} \chi_r \mu_0(x) w_{s,\theta} \cdot \nabla \Lambda_{s,\theta} dx \\ &= - \int_{\mathbb{R}^d} \chi_r \Lambda_{s,\theta} (\mu_0(x) \operatorname{div} w_{s,\theta} + \nabla \mu_0(x) \cdot w_{s,\theta}) dx - \int_{\mathbb{R}^d} \Lambda_{s,\theta} \mu_0(x) w_{s,\theta} \cdot \nabla \chi_r dx. \end{aligned} \quad (4.39)$$

The hypothesis $\|\nabla^3 G\|_\infty < \infty$ enters at this point only: it makes $\int_{\mathbb{R}^d} \mathbb{E}_0 [|w_{s,\theta}| |\nabla \Lambda_{s,\theta}|] \mu_0(x) dx$ finite, so that the left side of (4.39), after $\mathbb{E}_0$, converges as $r \rightarrow \infty$. The last integral of (4.39) is the term the cutoff adds; after the expectation over the path, (4.34), (4.38) and the Cauchy–Schwarz inequality bound it by

$$\begin{aligned} \frac{C}{r} \int_{\mathbb{R}^d} \mathbb{E}_0 [| \Lambda_{s,\theta} | |w_{s,\theta}|] \mu_0(x) dx &\leq \frac{C}{r} \|\nabla^2 G\|_\infty \int_{\mathbb{R}^d} \mathbb{E}_0 [|v_\theta|^2 |\Delta Z_0|] \mu_0(x) dx \\ &\leq \frac{C}{r} h^3 \|\nabla^2 G\|_\infty (\mathbb{E}(1 + |Z_0|)^{12})^{1/2}, \end{aligned}$$

so it drops out as $r \rightarrow \infty$, and (4.39) after $\mathbb{E}_0$ leaves

$$\int_{\mathbb{R}^d} \mathbb{E}_0 [w_{s,\theta} \cdot \nabla \Lambda_{s,\theta}] \mu_0(x) dx = - \int_{\mathbb{R}^d} \mathbb{E}_0 [\Lambda_{s,\theta} (\mu_0(x) \operatorname{div} w_{s,\theta} + \nabla \mu_0(x) \cdot w_{s,\theta})] dx. \quad (4.40)$$

Taking the expectation of (4.36), integrating against μ_0 and inserting (4.40) with $|\operatorname{div} w_{s,\theta}| \leq d \|\partial_x w_{s,\theta}\|$ gives, for fixed s and θ ,

$$\begin{aligned} & \left| \int_{\mathbb{R}^d} \mathbb{E}_0 \left[\left\langle \nabla^3 G(x + s v_\theta) [v_\theta^{\otimes 2}], \Delta Z_0 \right\rangle \right] \mu_0(x) dx \right| \\ &\leq \int_{\mathbb{R}^d} \mathbb{E}_0 \left[d |\Lambda_{s,\theta}| \|\partial_x w_{s,\theta}\| + 2 \|\nabla^2 G\|_\infty \|\partial_x v_\theta\| |w_{s,\theta}| |v_\theta| \right] \mu_0(x) dx + \int_{\mathbb{R}^d} \mathbb{E}_0 [| \Lambda_{s,\theta} | |w_{s,\theta}|] |\nabla \mu_0(x)| dx, \end{aligned}$$

and (4.34) with (4.38) bounds both expectations by $Ch^3 \|\nabla^2 G\|_\infty (1 + |x|)^6$. Integrating over s and θ with the weight $1 - s$ of (4.30),

$$\begin{aligned} \left| \int_{\mathbb{R}^d} \mathbb{E}_0 [\rho_0^G] \mu_0(x) dx \right| &\leq Ch^3 \|\nabla^2 G\|_\infty \int_{\mathbb{R}^d} (1 + |x|)^6 (\mu_0(x) + |\nabla \mu_0(x)|) dx \\ &\leq Ch^3 \|\nabla^2 G\|_\infty \left(1 + \sqrt{I(\mu_0)} \right) (\mathbb{E}(1 + |Z_0|)^{12})^{1/2}, \end{aligned} \quad (4.41)$$

the integral against $|\nabla \mu_0|$ bounded by the Cauchy–Schwarz inequality,

$$\int_{\mathbb{R}^d} (1 + |x|)^6 |\nabla \mu_0(x)| dx = \int_{\mathbb{R}^d} (1 + |x|)^6 |\nabla \log \mu_0(x)| \mu_0(x) dx \leq \sqrt{I(\mu_0)} (\mathbb{E}(1 + |Z_0|)^{12})^{1/2}.$$

By (3.25) at $k = 0$, adding (4.29) and (4.41) bounds (4.27) by the right side of (4.2). $\square$

4.3. Proof of Lemma 9

By the step map structure (4.6), μ_{k+1} is the pushforward of μ_k by the step map, convolved with the Gaussian, then averaged over the stage noise. Along that recursion (4.45) the pushforward raises the

Fisher information (4.42), the convolution lowers it through Stam's inequality (4.44), and the average does not increase it (4.43).

Three useful inequalities about the Fisher information (4.1) enter. The first one is the Fisher transport inequality. Let μ be a positive C^1 probability density, $\mathcal{T}$ a C^2 diffeomorphism with positive Jacobian determinant and $\mathcal{T}_\# \mu$ the law of $\mathcal{T}(x)$ for $x \sim \mu$. The density of $\mathcal{T}_\# \mu$ is $\mu(\mathcal{T}^{-1}(y))/\det \partial_x \mathcal{T}(\mathcal{T}^{-1}(y))$, and thus its score at $y = \mathcal{T}(x)$ is

$$\nabla \log(\mathcal{T}_\# \mu)(\mathcal{T}(x)) = (\partial_x \mathcal{T}(x))^{-\top} (\nabla \log \mu(x) - \nabla \log \det \partial_x \mathcal{T}(x)).$$

Whenever

$$\|\partial_x \mathcal{T} - I_d\|_\infty + \|\nabla \log \det \partial_x \mathcal{T}\|_\infty \leq Ch \leq \frac{1}{2},$$

the inverse Jacobian obeys $\|(\partial_x \mathcal{T})^{-1}\|_\infty \leq (1 - Ch)^{-1} \leq 1 + 2Ch$, so the score satisfies

$$\begin{aligned} |\nabla \log(\mathcal{T}_\# \mu)(\mathcal{T}(x))|^2 &\leq (1 + 2Ch)^2 (|\nabla \log \mu(x)| + Ch)^2 \\ &\leq (1 + 2Ch)^2 \left((1 + h) |\nabla \log \mu(x)|^2 + C^2 h (1 + h) \right), \end{aligned}$$

and the change of variables $y = \mathcal{T}(x)$ in the integral defining $I(\mathcal{T}_\# \mu)$ gives

$$I(\mathcal{T}_\# \mu) = \int_{\mathbb{R}^d} |\nabla \log(\mathcal{T}_\# \mu)(\mathcal{T}(x))|^2 \mu(x) dx \leq (1 + Ch) I(\mu) + Ch. \quad (4.42)$$

The second is that Fisher information is convex under mixtures. Let $\mu = \int \mu_a \mathcal{N}(0, \frac{2h}{3} I_d)(da)$ be a mixture of positive C^1 laws μ_a . Its score

$$\nabla \log \mu(x) = \int_{\mathbb{R}^d} \nabla \log \mu_a(x) \frac{\mu_a(x)}{\mu(x)} \mathcal{N}\left(0, \frac{2h}{3} I_d\right)(da)$$

is a conditional expectation of the scores of the components. Jensen's inequality then gives

$$I(\mu) \leq \int_{\mathbb{R}^d} I(\mu_a) \mathcal{N}\left(0, \frac{2h}{3} I_d\right)(da). \quad (4.43)$$

Finally, Stam's inequality (Blachman, 1965; Stam, 1959) adds the reciprocals over independent summands; since $\mathcal{N}(0, \frac{h}{2} I_d)$ has Fisher information $2d/h$, every probability measure μ satisfies

$$\frac{1}{I(\mu * \mathcal{N}(0, \frac{h}{2} I_d))} \geq \frac{1}{I(\mu)} + \frac{h}{2d}, \quad (4.44)$$

so that $I(\mu * \mathcal{N}(0, \frac{h}{2} I_d)) \leq 2d/h$ whatever $I(\mu)$, including $I(\mu) = \infty$.

Proof of Lemma 9 Let β be the constant of (4.42) for the maps $\mathcal{T}_{h,a}$; by (4.8) it depends only on M and d . Set $\delta := \beta + d^{-1}$, and let h_0 be the smallest of the threshold of Lemma 10, $\frac{1}{2}$ and δ^{-1} . At step k ,

(4.6) reads $Z_{k+1} = \mathcal{T}_{h,A_k}(Z_k) + \sqrt{h/2} \eta_k$ with A_k and η_k the stage noise (3.12) and the Gaussian (4.5) of the increments over $[kh, (k+1)h]$. These increments are independent of $\mathcal{F}_k$ (Section 3.2), and η_k is independent of A_k , so that

$$\mu_{k+1} = \int_{\mathbb{R}^d} ((\mathcal{T}_{h,a})_{\#} \mu_k) * \mathcal{N}\left(0, \frac{h}{2} I_d\right) \mathcal{N}\left(0, \frac{2h}{3} I_d\right) (da), \quad (4.45)$$

the outer measure being the law of A_k . Every μ_{k+1} is a mixture of Gaussians of covariance $\frac{h}{2} I_d$, hence positive and C^∞ , and (4.44) with (4.43) gives $I(\mu_1) \leq 2d/h$. For $k \geq 1$ and every a , (4.44) and then (4.42) with (4.8) give

$$\frac{1}{I(((\mathcal{T}_{h,a})_{\#} \mu_k) * \mathcal{N}(0, \frac{h}{2} I_d))} \geq \frac{h}{2d} + \frac{1}{I((\mathcal{T}_{h,a})_{\#} \mu_k)} \geq \frac{h}{2d} + \frac{1}{(1 + \beta h)I(\mu_k) + \beta h}.$$

This bound does not depend on a , and (4.43) over the outer measure of (4.45) gives

$$\frac{1}{I(\mu_{k+1})} \geq \frac{h}{2d} + \frac{1}{(1 + \beta h)I(\mu_k) + \beta h}. \quad (4.46)$$

Compare with the sequence

$$\bar{I}_1 := \frac{2d}{h}, \quad \frac{1}{\bar{I}_{k+1}} := \frac{h}{2d} + \frac{1}{(1 + \delta h)\bar{I}_k}. \quad (4.47)$$

The right side of the recursion in (4.47) is positive and decreases in $\bar{I}_k$, so $\bar{I}_{k+1}$ increases in $\bar{I}_k$. The recursion has the fixed point $2d\delta/(1 + \delta h)$, and $\bar{I}_{k+1} \leq \bar{I}_k$ exactly when $\bar{I}_k \geq 2d\delta/(1 + \delta h)$; since $\bar{I}_1 = 2d/h$ satisfies this, induction gives

$$\bar{I}_1 \geq \bar{I}_2 \geq \dots \geq \frac{2d\delta}{1 + \delta h} \geq d\delta,$$

the last step since $\delta h \leq 1$. Since $(\delta - \beta)\delta \geq \beta/d$ by the choice of δ , this gives $\bar{I}_k \geq \beta/(\delta - \beta)$, hence

$$(1 + \beta h)\bar{I}_k + \beta h \leq (1 + \delta h)\bar{I}_k, \quad k \geq 1. \quad (4.48)$$

Whenever $I(\mu_k) \leq \bar{I}_k$, (4.46), (4.48) and (4.47) give, in this order,

$$\frac{1}{I(\mu_{k+1})} \geq \frac{h}{2d} + \frac{1}{(1 + \beta h)\bar{I}_k + \beta h} \geq \frac{h}{2d} + \frac{1}{(1 + \delta h)\bar{I}_k} = \frac{1}{\bar{I}_{k+1}};$$

induction from $I(\mu_1) \leq \bar{I}_1$ gives $I(\mu_k) \leq \bar{I}_k$ for every $k \geq 1$. Iterating (4.47),

$$\frac{1}{\bar{I}_k} = \frac{h}{2d} \sum_{j=0}^{k-1} (1 + \delta h)^{-j},$$

where $(1 + \delta h)^{-j} \geq e^{-\delta jh} \geq e^{-\delta}$ for $jh \leq 1$. For $kh \leq 1$ all k summands satisfy $jh \leq 1$, and for $kh > 1$ so do the $\lfloor 1/h \rfloor \geq (2h)^{-1}$ summands with $j < \lfloor 1/h \rfloor$, since $h \leq \frac{1}{2}$; hence

$$\bar{I}_k \leq \frac{2de^\delta}{kh} \quad \text{for } kh \leq 1, \quad \bar{I}_k \leq 4de^\delta \quad \text{for } kh > 1,$$

and $I(\mu_k) \leq \bar{I}_k$ gives (4.3) with $C = 4de^\delta$. $\square$

5. Proof of Theorem 1

Theorem 1 follows by summing the bound (4.4) over the steps. The lemma below carries out the summation, making the finiteness of $\int_0^1 (t(1-t))^{-1/2} dt$ quantitative for the discrete sum with its exponential weight. The first step, where μ_0 is a point mass and Lemma 8 does not apply, and the last step, where $G_{N-1} = g$ is propagated over no time, are treated separately in the proof of the theorem.

Lemma 11 *Let λ be the rate of Lemma 2. There is C depending only on λ such that for every $h > 0$ with $\lambda h \leq 1$ and every $n \geq 2$,*

$$h \sum_{k=1}^{n-1} (1 + (kh)^{-1/2}) (1 + ((n-k)h)^{-1/2}) e^{-\lambda(n-k)h} \leq C. \quad (5.1)$$

Proof of Lemma 11 Write $i := n - k$ and expand the product into four sums. Since $1 - e^{-\lambda h} \geq \frac{\lambda h}{2}$ for $\lambda h \leq 1$, and since $t \mapsto t^{-1/2}$ decreases,

$$\begin{aligned} h \sum_{i \geq 1} e^{-\lambda i h} &\leq \frac{h}{1 - e^{-\lambda h}} \leq \frac{2}{\lambda}, \\ h \sum_{i \geq 1} (ih)^{-1/2} e^{-\lambda i h} &\leq h \sum_{ih \leq 1} (ih)^{-1/2} + h \sum_{ih > 1} e^{-\lambda i h} \leq \int_0^1 t^{-1/2} dt + \frac{2}{\lambda} = 2 + \frac{2}{\lambda}, \end{aligned}$$

which bounds the sums with no factor $(kh)^{-1/2}$. For the sum carrying $(kh)^{-1/2}$ alone, split at $k = n/2$:

$$h \sum_{k \leq n/2} (kh)^{-1/2} e^{-\lambda(n-k)h} \leq 2\sqrt{nh} e^{-\lambda nh/2}, \quad h \sum_{k > n/2} (kh)^{-1/2} e^{-\lambda(n-k)h} \leq \left(\frac{nh}{2}\right)^{-1/2} \min\left(\frac{nh}{2}, \frac{2}{\lambda}\right),$$

the sum over $k \leq n/2$ at most $2(e\lambda)^{-1/2}$ since $\sqrt{t} e^{-\lambda t/2} \leq (e\lambda)^{-1/2}$ for $t \geq 0$, and the sum over $k > n/2$ at most $(2/\lambda)^{1/2}$ in either case of the minimum. For the sum carrying both $(kh)^{-1/2}$ and $((n-k)h)^{-1/2}$ the exponential is dropped:

$$h \sum_{k=1}^{n-1} (kh)^{-1/2} ((n-k)h)^{-1/2} = \sum_{k=1}^{n-1} \frac{1}{\sqrt{k(n-k)}} \leq 2 \sum_{k \leq n/2} \frac{1}{\sqrt{kn/2}} \leq 4.$$

Adding the four bounds proves (5.1). $\square$

Proof of Theorem 1 In this proof C depends on m, M, R and d . Let h_0 be the smallest of 1, λ^{-1} , the thresholds of Lemmas 8 and 9, and those of Lemma 6 with $q = 2$ and $q = 6$.

By the duality (2.1) it suffices to bound $|\mathbb{E}g(Z_N) - \mathbb{E}g(X_{N_h})|$ uniformly over $g \in C^\infty(\mathbb{R}^d)$ with all derivatives bounded and $\|\nabla g\|_\infty \leq 1$, since the mollifications of a 1-Lipschitz function are of this class and converge to it uniformly. Both integrals of (2.1) are then defined, the measures compared having finite first moments, μ_N by Lemma 6 and ν_N by the linear growth $|\nabla U(x)| \leq M(1 + |x|)$.

With $G_k = P_{(N-k-1)h}g$ and $i = N - k - 1$ as in (4.4), splitting off the last step of the telescoping sum (3.28) gives

$$\mathbb{E}[g(Z_N)] - \mathbb{E}[g(X_{Nh})] = \mathbb{E}[g(Z_N) - g(Z_{N-1}(h))] + \sum_{k=0}^{N-2} \mathbb{E}\left[\mathbb{E}_k[G_k(Z_{k+1}) - G_k(Z_k(h))]\right]. \quad (5.2)$$

At the last step $k = N - 1$, $G_{N-1} = g$ is 1-Lipschitz; conditioning on $\mathcal{F}_{N-1}$, (3.23) and then Lemma 6 at $q = 2$ give

$$\left|\mathbb{E}[g(Z_N) - g(Z_{N-1}(h))]\right| \leq \mathbb{E}|\Delta Z_{N-1}| \leq Ch^2 \mathbb{E}(1 + |Z_{N-1}|)^4 \leq Ch^2(1 + |Z_0|)^4. \quad (5.3)$$

For $0 \leq k \leq N - 2$ the propagation time is $ih \geq h$, and (3.4) with (3.6) gives

$$\|\nabla G_k\|_\infty + \|\nabla^2 G_k\|_\infty \leq C(1 + (ih)^{-1/2})e^{-\lambda ih}, \quad \|\nabla^3 G_k\|_\infty \leq C \min(1, ih)^{-1}e^{-\lambda ih}. \quad (5.4)$$

At the first step $k = 0$, present when $N \geq 2$, the state Z_0 is deterministic and $i = N - 1 \geq 1$, so (5.4) with $(ih)^{-1/2} \leq h^{-1/2} \leq h^{-1}$ and $\min(1, ih)^{-1} \leq h^{-1}$ gives

$$\|\nabla G_0\|_\infty + \|\nabla^2 G_0\|_\infty + \|\nabla^3 G_0\|_\infty \leq Ch^{-1},$$

and Lemma 7 gives

$$\left|\mathbb{E}[G_0(Z_1) - G_0(Z_0(h))]\right| \leq Ch^3 \cdot h^{-1}(1 + |Z_0|)^6 = Ch^2(1 + |Z_0|)^6. \quad (5.5)$$

For $1 \leq k \leq N - 2$, the increments over the step are independent of $\mathcal{F}_k$, so the summand of (5.2) is the left side of (4.2) with μ_k in place of μ_0 and $G = G_k$. Lemma 9 gives μ_k a positive C^∞ density with $\sqrt{I(\mu_k)} \leq C(1 + (kh)^{-1/2})$, Lemma 6 with $q = 6$ gives $(\mathbb{E}(1 + |Z_k|)^{12})^{1/2} \leq C(1 + |Z_0|)^6$, and G_k has bounded derivatives up to third order by (5.4). Lemma 8 and then (5.4) bound the summand by

$$\begin{aligned} & Ch^3 (\|\nabla G_k\|_\infty + \|\nabla^2 G_k\|_\infty) \left(1 + \sqrt{I(\mu_k)}\right) (\mathbb{E}(1 + |Z_k|)^{12})^{1/2} \\ & \leq C(1 + |Z_0|)^6 h^3 (1 + (kh)^{-1/2}) (1 + (ih)^{-1/2}) e^{-\lambda ih}, \end{aligned}$$

and, for $N \geq 3$ (for $N \leq 2$ the sum is empty), Lemma 11 with $n = N - 1$, for which $i = n - k$, gives

$$h \sum_{k=1}^{N-2} (1 + (kh)^{-1/2}) (1 + (ih)^{-1/2}) e^{-\lambda ih} \leq C,$$

so that

$$\sum_{k=1}^{N-2} \left|\mathbb{E}[G_k(Z_{k+1}) - G_k(Z_k(h))]\right| \leq C(1 + |Z_0|)^6 h^2. \quad (5.6)$$

Inserting (5.3), (5.5) and (5.6) into (5.2),

$$\left|\mathbb{E}g(Z_N) - \mathbb{E}g(X_{Nh})\right| \leq C(1 + |Z_0|)^6 h^2,$$

which depends on neither g nor N . Taking the supremum over g proves (2.4).

For (2.5) and the same class of g , (3.5) gives $|(P_{Nh}g)(Z_0) - \pi(g)| \leq Ce^{-\lambda Nh}(1 + |Z_0|)$, and $(P_{Nh}g)(Z_0) = \mathbb{E}g(X_{Nh})$, so (2.1) gives

$$\mathcal{W}_1(v_N, \pi) \leq C(1 + |Z_0|) e^{-\lambda Nh}.$$

The triangle inequality for $\mathcal{W}_1$ then gives (2.5). $\square$

6. Sharpness of the second order

Proof of Theorem 2 With $\nabla U(x) = x$, the stage of SRK-I in (2.2) is

$$H_k = \left(1 - \frac{3h}{4}\right)Z_k + \frac{3\sqrt{2}}{2h} \int_{kh}^{(k+1)h} (B_s - B_{kh}) ds,$$

and the update becomes

$$Z_{k+1} = \left(1 - h + \frac{h^2}{2}\right)Z_k + \sqrt{2}(B_{(k+1)h} - B_{kh}) - \sqrt{2} \int_{kh}^{(k+1)h} (B_s - B_{kh}) ds. \quad (6.1)$$

For SRK-II, (2.3) gives

$$\nabla U(H_k^+) + \nabla U(H_k^-) = (2-h)Z_k + \frac{2\sqrt{2}}{h} \int_{kh}^{(k+1)h} (B_s - B_{kh}) ds,$$

and the update is (6.1) again. The noise terms of (6.1) are independent over k and centered Gaussian, with variance

$$2 \operatorname{Var}\left(B_h - \int_0^h B_s ds\right) = 2\left(h - h^2 + \frac{h^3}{3}\right),$$

since $\operatorname{Var}(B_h) = h$, $\operatorname{Var}\left(\int_0^h B_s ds\right) = \frac{h^3}{3}$ and $\operatorname{Cov}(B_h, \int_0^h B_s ds) = \frac{h^2}{2}$. For $h \in (0, 1]$ the coefficient $1 - h + \frac{h^2}{2} = \frac{1}{2}(1 + (1-h)^2)$ lies in $[\frac{1}{2}, 1)$, so (6.1) is a Gaussian autoregression with the invariant law $\pi_h = \mathcal{N}(0, \sigma_h^2)$, where

$$\sigma_h^2 = \frac{2h - 2h^2 + \frac{2}{3}h^3}{1 - (1 - h + \frac{h^2}{2})^2} = \frac{1 - h + \frac{1}{3}h^2}{1 - h + \frac{1}{2}h^2 - \frac{1}{8}h^3} = 1 - \frac{h^2}{6} + \mathcal{O}(h^3). \quad (6.2)$$

The solution Z_k of (6.1) is Gaussian, with mean $(1 - h + \frac{h^2}{2})^k Z_0$ and variance $\sigma_h^2(1 - (1 - h + \frac{h^2}{2})^{2k})$. Its law therefore converges to π_h . Let ξ be a standard Gaussian, so that $\sigma_h \xi$ has law π_h and ξ has law π . Every $f \in \operatorname{Lip}_1(\mathbb{R})$ satisfies

$$|\mathbb{E}f(\sigma_h \xi) - \mathbb{E}f(\xi)| \leq \mathbb{E}|\sigma_h \xi - \xi| = |\sigma_h - 1| \mathbb{E}|\xi|,$$

with equality at $f(x) = |x|$. Hence (2.1) gives

$$\mathcal{W}_1(\mathcal{N}(0, \sigma_h^2), \mathcal{N}(0, 1)) = \sqrt{\frac{2}{\pi}} |\sigma_h - 1|,$$

and $\sigma_h = 1 - \frac{h^2}{12} + \mathcal{O}(h^3)$ by (6.2) gives (2.6).

For $f(x) = \cos x$, $\pi_h(f) = e^{-\sigma_h^2/2}$ and $\pi(f) = e^{-1/2}$, so by (6.2)

$$\pi_h(f) - \pi(f) = e^{-1/2}(e^{(1-\sigma_h^2)/2} - 1) = \frac{e^{-1/2}}{12} h^2 + \mathcal{O}(h^3).$$

Since Z_k converges in law to π_h and f is bounded and continuous, $\mathbb{E}f(Z_k) \rightarrow \pi_h(f)$, hence $\mathbb{E}\left[\frac{1}{N} \sum_{k=0}^{N-1} f(Z_k)\right] \rightarrow \pi_h(f)$, and Jensen's inequality in (1.2) gives

$$\liminf_{N \rightarrow \infty} \operatorname{MSE}(N, h) \geq (\pi_h(f) - \pi(f))^2 = \frac{e^{-1}}{144} h^4 + \mathcal{O}(h^5),$$

which is (2.7). $\square$

7. Conclusion

For the two stochastic Runge–Kutta integrators of Yang & Wang (2026), the uniform-in-time Wasserstein-1 bound $h^2(1 + \log \frac{1}{h})$ of Ye (2026) is sharpened to h^2 , and a Gaussian example shows that h^2 is the order of the bias of their invariant law and that the squared bias h^4 of their time averages is attained. The argument replaces the pointwise local weak error of one step by its average over the law of the chain, whose Fisher information the Gaussian part of each step controls; the control of that Fisher information uses nothing about the integrator beyond the structure (4.6) of its step.

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